OsdagBridge

Open Source Software for Steel Girder Bridge Design

Design Report

Project Name	baseline1
Project Location	Pune, Maharashtra
Author / Designer	
Reviewer	
Organization	
Client Name and Organization	
Job Number	
Date	2026-08-19
Report Version	

Table of Contents

Executive Summary

This section provides a concise summary of the bridge design, key inputs, governing loads, and final design outcomes.

Project Overview

Bridge Type	Highway Bridge
Design Standard	IRC 5, IRC 6, IRC 22, IRC 24, IS 800
Span	20 m
Carriageway Width	4.25 m
No. of Girders	3
Girder Spacing	1.7167
Deck Thickness	250.0
Overall Design Status	Fail (Fatigue Normal Stress)
Governing Check	Fatigue Normal Stress
Overall Utilization Ratio (max)	1.35 (Girder)

[PLACEHOLDER: Figure 1 – Overall Bridge Plan]

Figure 1 – Overall Bridge Plan

[PLACEHOLDER: Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)]

Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)

[PLACEHOLDER: Figure 3 – 3D View of Bridge Superstructure]

Figure 3 – 3D View of Bridge Superstructure

Table 1 – Final Bridge Geometry (after optimization)

	G1	**G2**	**G3**
Member ID	G1M1	G2M1	G3M1
Section Designation	1120 x 340 x 16 x 340 x 16	1120 x 340 x 16 x 340 x 16	1120 x 340 x 16 x 340 x 16
Governing Check	Fatigue Normal Stress	Fatigue Normal Stress	Fatigue Normal Stress
Utilization Ratio	1.35 (Girder)	1.35 (Girder)	1.35 (Girder)

Note: Utilization ratio (UR) = demand / capacity. A value < 1.0 *indicates a passing check.*

Key Design Outcomes Summary

Girder design pass
Cross bracing design pass
End Diaphragm design pass
Deck design pass

Design Assumptions and Limitations

- Additional inputs not provided by the user were assumed by software per IRC/IS code defaults or practical consideration.
- Grillage analysis was performed using OSPGrillage assuming simply supported I-girders.
- Substructure and foundation design are not included in this report.
- Splice connections and bearings are not designed in this version.

1 Project Information

This section records all project metadata as entered by the designer.

1.1 Project and Design Team Details

Project Name	baseline1
Project Location	Pune, Maharashtra
Designer	
Reviewer	
Organization	
Client	
Software Version	OsdagBridge

1.2 Applicable Codes and Standards

- Indian Roads Congress (IRC) 5: General Features of Design
- Indian Roads Congress (IRC) 6: Loads and Load Combinations
- Indian Roads Congress (IRC) 22: Composite Construction (Limit State Design)
- Indian Roads Congress (IRC) 24: Steel Road Bridges (Limit State Method)
- Indian Roads Congress (IRC) 112: Concrete Road Bridges (deck design)
- Indian Roads Congress Special Publication (IRC SP) 114: Seismic Design of Road Bridges
- Indian Standard (IS) 800: General Construction in Steel
- Indian Standard (IS) 2062: Hot Rolled Structural Steel Specification
- Indian Standard (IS) 6006: Steel Bearings

2 Input Parameters

This section documents all inputs provided to OsdagBridge. User-provided inputs are clearly distinguished from software-assumed defaults. Where the user did not supply a value, the software has applied the IRC/IS code default or an empirical guideline; these are annotated with an asterisk (*).

2.1 Basic Inputs (User-Defined)

Note: These inputs are mandatory and were provided by the user.

Table 2.1: Project Location

parameter	value
Project Location	Pune, Maharashtra
Latitude / Longitude	18.5204, 73.8567
Seismic Zone (IRC 6)	III
Basic Wind Speed (IRC 6)	39 m/s
Shade Temp. Max / Min (IRC 6)	43.3 °C / 1.7 °C

Table 2.2: Bridge Geometry

parameter	value
Type of Structure	Highway Bridge
Span (m)	20 m
Carriageway Width (m)	4.25 m
Include Median	No
Footpath	None
Skew Angle (degrees)	(IRC 24 Cl. 504.8 limit: ±15°)

Table 2.3: Material Selection

parameter	value
Girder Steel Grade (IS 2062)	E 350A
Cross Bracing Steel Grade	E 350A
End Diaphragm Steel Grade	E 350A
Concrete Deck Grade (IRC 22)	M40

** Software default value*

2.2 Additional Inputs

Where the user has modified additional inputs, those values are reported here. Where no modification was made, the software default is shown.

Table 2.4: Typical Section Details

parameter	value
Overall Bridge Width (m)	5.15
No. of Girders	3
Girder Spacing (m)	1.7167 m
Deck Overhang Width (m)	0.8583 m
Deck Thickness (mm)	250.0 mm
Footpath Width (m)	0.0 m (IRC 5 Cl. 104.3.6 min: 1.5 m)
No. of Traffic Lanes	(per IRC 5 Cl. 104.3.1)

Table 2.5: Components Details

parameter	value
Crash Barrier Type	IRC 5 - RCC Crash Barrier
Crash Barrier Load (kN/m)	7.56
Railing Type	IRC 5 - RCC Railing
Railing Load (kN/m)	6.806
Wearing Course Material	Concrete
Wearing Course Thickness (mm)	50.0 mm

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G1	G1M1	Optimized	Welded	Girder Symmetric
G2	G2M1	Optimized	Welded	Girder Symmetric
G3	G3M1	Optimized	Welded	Girder Symmetric

Table 2.7: Girder Section Dimensions

Girder	Total Depth, D (mm)	Web, t_w (mm)	Top Flange (b_{tf}, t_{tf}) mm	Bottom Flange (b_{bf}, t_{bf}) mm
G1	1.12 mm	0.008 mm	0.34 mm, 0.016 mm	0.34 mm, 0.016 mm
G2	1.12 mm	0.008 mm	0.34 mm, 0.016 mm	0.34 mm, 0.016 mm
G3	1.12 mm	0.008 mm	0.34 mm, 0.016 mm	0.34 mm, 0.016 mm

Table 2.8: Girder Restraint and Stiffener Details

Girder	Torsional / Warping Restraint	Web Philosophy	Intermediate Stiffeners	Longitudinal Stiffeners	Bearing Stiffener
G1	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 1635 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G2	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 1635 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G3	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 1635 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm

Table 2.9: Member Properties: Cross Bracing Details

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)
G1 to G2	B1M1 to B1M1	X-Bracing	IS 100 x 100 x 10	10.0 m
G2 to G3	B2M1 to B2M1	X-Bracing	IS 100 x 100 x 10	10.0 m

Table 2.10: Member Properties: End Diaphragm Details

Location	Member IDs	Type of Bracing	Bracing Section
G1 to G2	E1M1 to E1M2	Cross Bracing	IS 100 x 100 x 10
G2 to G3	E2M1 to E2M2	Cross Bracing	IS 100 x 100 x 10

Table 2.11: Shear Connector Details

parameter	value
Stud Diameter (mm)	20.0 mm
Stud Height (mm)	100.0 mm
Stud f_y (MPa)	385.0 MPa
Stud f_u (MPa)	495.0 MPa
No. of Studs per Section	2

Note: All values are per IRC 22 Table 1 unless user-modified.

Table 2.12: Partial Safety Factors

parameter	**value**
γ_{M0} (Yielding / Buckling)	1.1
γ_{M1} (Ultimate Stress)	1.25
γ_{C} (Concrete, Basic)	1.5
γ_{s} (Reinforcement)	1.15
γ_{v} (Shear Connectors)	1.25
γ_{fft} (Fatigue Load)	1.0
γ_{Mft} (Fatigue Strength)	1.35

3 Loads and Load Combinations

This section summarizes all loads applied to the bridge and the load combinations considered for analysis and design.

Table 3.1: Dead Load – Self Weight

parameter	value
Steel Self-Weight Applied	76.4 kN/m^3
Concrete Deck Weight	24.5 kN/m^3
Self-Weight Factor	1.0

Table 3.2: Dead Load for Surfacing (DW)

parameter	value
Wearing Course Load	Concrete x 50.0
Additional SIDL (Crash Barrier)	7.56 kN/m per barrier
Railing Load	6.806 kN/m*

Table 3.3: Live Loads (LL)

parameter	value
Vehicles Considered	Class A, Class 70R (Wheeled), Class 70R (Tracked), Class AA (Wheeled), Class AA (Tracked), Class SV, Class 70R (Bogie), Class Fatigue
Impact Factor (IRC 6)	Class A: 1.269, Class AA/70R: 1.100
Braking Load (IRC 6)	N/A
Footpath Live Load (if applicable)	

* *Software default value*

Table 3.4: Wind Load (WL) — per IRC 6

parameter	value
Basic Wind Speed, Vb	39 m/s [from Project Location]
Terrain Type	Plain Terrain
Average Exposed Height, H (m)	10 m
Hourly Mean Wind Speed, Vz	
Hourly Wind Pressure, Pz	
Transverse Wind Force	
Longitudinal Wind Force	
Vertical Wind Force	

Table 3.5: Earthquake Load (EL) — per IRC 6

parameter	value
Seismic Zone	III [from Project Location]
Zone Factor, Z	0.16
Importance Factor, I	1.0
Type of Soil	Type I – Rocky or Hard
Sa/g	2.8
Horizontal Seismic Coefficient, Ah	0.224
Vertical Seismic Coefficient, Av	0.1493
Horizontal Seismic Force (longitudinal)	kN
Horizontal Seismic Force (transverse)	kN

Table 3.6: Temperature Load (TL) — per IRC 6

parameter	value
Maximum Shade Temperature	43.3 °C
Minimum Shade Temperature	1.7 °C
Effective Bridge Temp. Range	-8.30 to 58.30 °C
Temperature Rise / Fall for Design	+33.30 °C / −33.30 °C

Table 3.7: Load Combinations

Combination ID	Load Cases
ULS-01	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.50) + WL(0.90) + TL(0.90)
ULS-02	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(1.50) + TL(0.90)
ULS-03	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(0.90) + TL(1.50)
ULS-04	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + VC(1.00)
ULS-05	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + BI(1.00)
ULS-06	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + FB(1.00)
ULS-07	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(1.50)
ULS-08	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(0.75)
SLS-01	DL(1.00) + SIDL(1.20/1.00) + LL(1.00) + WL(0.60) + TL(0.60)
SLS-02	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(1.00) + TL(0.60)
SLS-03	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.60) + TL(1.00)
SLS-04	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.50) + TL(0.50)
SLS-05	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.60) + TL(0.50)

SLS-06	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.50) + TL(0.60)
SLS-07	DL(1.00) + SIDL(1.20/1.00) + LL(0.00) + WL(0.00) + TL(0.50)

Note: All IRC 6 load combinations are auto-generated by OsdagBridge. User-defined custom combinations, if any, are appended.

4 Analysis Results

A grillage model was used for structural analysis. The deck is idealized as a grid of elastic beam elements — longitudinal members represent the composite steel girders with effective slab, and transverse members represent the slab or cross frames. This section summarizes the critical output from that analysis.

Table 4.1: Summary of Maximum Demands

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
BASIC_2: 1.35DL + 1.75DW + 1.15LL + 1.5WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
1.0 DL	364.982	G1	10.769	83.323	G1	18.462	-234.692	234.692
SLS_RARE_3: 1.0DL + 1.2DW + 0.75LL + 0.6WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
Envelope ULS	690.518	G1	9.231	160.937	G1	0.000	-452.633	409.054
BASIC_6: 1.0DL + 1.0DW + 1.15LL + 0.9WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
Crash barrier load	83.546	G1	10.769	19.038	G1	18.462	-48.161	52.081
DD	328.505	G1	10.769	75.204	G1	18.462	-211.616	211.616
EQ_Z	0.000	—	—	0.000	—	—	0.000	-0.000
ACCIDENTAL_3: 1.0DL + 1.0DW + 0.75LL	690.518	G1	9.231	160.937	G1	0.000	-452.633	409.054
SEISMIC_4: 1.0DL + 1.0DW + 0.2LL + 0.75EL	626.670	G1	9.231	146.678	G1	0.000	-412.327	368.749
EQ_X	0.000	—	—	0.000	—	—	0.000	-0.000
SEISMIC_1: 1.35DL + 1.75DW + 0.2LL + 1.5EL	626.670	G1	9.231	146.678	G1	0.000	-412.327	368.749
SLS_QP_1: 1.0DL + 1.2DW	430.683	G1	9.231	98.364	G1	0.000	-277.015	277.015
SLS_RARE_5: 1.0DL + 1.0DW + 0.75LL + 1.0WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
WL Longitudinal	0.000	—	—	0.000	—	—	-0.000	0.000
1.5 EQ (c)	95.771	G1	10.769	21.389	G1	18.462	60.458	-60.458
WL Transverse	0.000	—	—	0.000	—	—	-0.000	0.000
SW	36.478	G1	10.769	8.119	G1	18.462	-23.076	23.076
BASIC_5: 1.0DL + 1.0DW + 1.15LL + 1.5WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
SLS_RARE_6: 1.0DL + 1.0DW + 0.75LL + 0.6WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
SLS_FREQUENT_4: 1.0DL + 1.0DW + 0.75LL + 0.5WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
WL Uplift	36.277	G1	10.769	8.102	G1	18.462	22.901	-22.901
1.0 DL + 1.0 LL	624.817	G1	9.231	145.896	G1	0.000	-410.309	366.731
BASIC_4: 1.0DL + 1.0DW + 1.5LL + 0.9WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
1.5 EQ (b)	28.731	G1	10.769	6.417	G1	18.462	18.137	-18.137
SLS_RARE_4: 1.0DL + 1.0DW + 1.0LL + 0.6WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
SEISMIC_3: 1.0DL + 1.0DW + 0.2LL + 1.5EL	626.670	G1	9.231	146.678	G1	0.000	-412.327	368.749
BASIC_3: 1.35DL + 1.75DW + 1.15LL + 0.9WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
SLS_RARE_2: 1.0DL + 1.2DW + 0.75LL + 1.0WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
SLS_FREQUENT_5: 1.0DL + 1.0DW + 0.2LL + 0.6WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
ACCIDENTAL_2: 1.0DL + 1.0DW + 0.75LL	690.518	G1	9.231	160.937	G1	0.000	-452.633	409.054
SLS_FREQUENT_2: 1.0DL + 1.2DW + 0.2LL + 0.6WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
Envelope SLS	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
SEISMIC_2: 1.35DL + 1.75DW + 0.2LL + 0.75EL	626.670	G1	9.231	146.678	G1	0.000	-412.327	368.749
SLS_FREQUENT_3: 1.0DL + 1.2DW + 0.2LL + 0.5WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
1.0 DW	65.701	G1	9.231	15.041	G1	0.000	-42.323	42.323
1.5 EQ (a)	28.731	G1	9.231	6.417	G1	0.000	18.137	-18.137
BASIC_1: 1.35DL + 1.75DW + 1.5LL + 0.9WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
SLS_QP_2: 1.0DL + 1.0DW	430.683	G1	10.769	98.364	G1	18.462	-277.015	277.015
EQ_Y	63.847	G1	9.231	14.259	G1	18.462	40.305	-40.305
SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
SLS_FREQUENT_6: 1.0DL + 1.0DW + 0.2LL + 0.5WL	654.241	G1	9.231	152.835	G1	0.000	-429.732	386.153
ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	690.518	G1	9.231	160.937	G1	0.000	-452.633	409.054
1.0 WL	36.277	G1	9.231	8.102	G1	18.462	22.901	-22.901

Table 4.2: Reactions at Supports

Load Case	Left Support (kN)	Right Support (kN)

Table 4.3: Deflection Summary (Live Load & Total Load)

parameter	value
Deflection due to Live Load, δ_{LL}	15.611 mm
Allowable Live Load Deflection (Δ_{allow})	L/800 = 25.0 mm
Live Load Deflection Check Status	PASS
Deflection due to Total Load, δ_{total}	37.125 mm
Allowable Total Deflection (Δ_{allow})	L/600 = 33.3 mm
Total Load Deflection Check Status	FAIL

[PLACEHOLDER: Bending Moment Envelope (Envelope ULS): Max/min BM along span. X-axis: distance from left support (m). Y-axis: Bending Moment (kN-m).]

[PLACEHOL X-axis: distan

[PLACEHOLDER: Vertical Deflection D_y (1.0 LL): Maximum deflection along span. Load Case: 1.0 LL, Combination: D_y. Nodes shown. Isometric view.]

5 Design Checks

This section presents all structural design checks performed by OsdagBridge. For each member, the demand from the governing load combination, the code-based capacity, and the utilization ratio are tabulated. All checks reference IS 800:2007 and IRC 22:2014 unless stated otherwise.

5.1 Plate Girder Design

Table 5.1: Girder Section Properties (Final Optimized / User-selected)

Girder	Property	Value
G1	Depth, D (mm)	1120.0
	Top Flange Width, b_f (mm)	340.0
	Bottom Flange Width, b_f (mm)	340.0
	Top Flange Thickness, t_f (mm)	16.0
	Bottom Flange Thickness, t_f (mm)	16.0
	Web Thickness, t_w (mm)	8.0
	Gross Area, A (cm^2)	195.84
	Moment of Inertia, I_z (cm^4)	417402.06
	Elastic Section Modulus, Z_{ez} (cm^3)	7453.61
	Plastic Section Modulus, Z_{pz} (cm^3)	8373.25
	Effective Slab Width, b_{eff} (mm)	1716.7
	Transformed Composite I_z (cm^4)	1131831.28
	Depth to Plastic Neutral Axis (mm)	250.4
G2	Depth, D (mm)	1120.0
	Top Flange Width, b_f (mm)	340.0
	Bottom Flange Width, b_f (mm)	340.0
	Top Flange Thickness, t_f (mm)	16.0
	Bottom Flange Thickness, t_f (mm)	16.0
	Web Thickness, t_w (mm)	8.0

	Gross Area, A (cm^2)	195.84
	Moment of Inertia, I_z (cm^4)	417402.06
	Elastic Section Modulus, Z_{ez} (cm^3)	7453.61
	Plastic Section Modulus, Z_{pz} (cm^3)	8373.25
	Effective Slab Width, b_{eff} (mm)	1716.7
	Transformed Composite I_z (cm^4)	1131831.28
	Depth to Plastic Neutral Axis (mm)	250.4
G3	Depth, D (mm)	1120.0
	Top Flange Width, b_f (mm)	340.0
	Bottom Flange Width, b_f (mm)	340.0
	Top Flange Thickness, t_f (mm)	16.0
	Bottom Flange Thickness, t_f (mm)	16.0
	Web Thickness, t_w (mm)	8.0
	Gross Area, A (cm^2)	195.84
	Moment of Inertia, I_z (cm^4)	417402.06
	Elastic Section Modulus, Z_{ez} (cm^3)	7453.61
	Plastic Section Modulus, Z_{pz} (cm^3)	8373.25
	Effective Slab Width, b_{eff} (mm)	1716.7
	Transformed Composite I_z (cm^4)	1131831.28
	Depth to Plastic Neutral Axis (mm)	250.4

Table 5.2: Girder Section Classification

	Element	Slenderness Ratio	Class Limit	Classification
G1	Top Flange	10.38	11.49	Semi-Compact
	Web	136.0	106.49	Slender
	Overall Section	—	—	Slender
G2	Top Flange	10.38	11.49	Semi-Compact
	Web	136.0	106.49	Slender
	Overall Section	—	—	Slender
G3	Top Flange	10.38	11.49	Semi-Compact
	Web	136.0	106.49	Slender
	Overall Section	—	—	Slender

Note: IS 800:2007 Table 2

Table 5.3: Moment Capacity Check

	Parameter	Formula	Value	Status
G1	Applied Moment, M_u	Governing LC (ULS)	697.45 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	4409.64 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.16	PASS
G2	Applied Moment, M_u	Governing LC (ULS)	697.45 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	4409.64 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.16	PASS
G3	Applied Moment, M_u	Governing LC (ULS)	697.45 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	4409.64 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.16	PASS

Note: IRC 22 Cl. 603.3.1, IS 800 Cl. 8.2.1

Table 5.4: Shear Capacity Check

	Parameter	Formula	Value	Status
G1	Applied Shear, V_u	Governing LC (ULS)	130.99 kN	—
	Shear Area, A_v	$d_w \times t_w$	8704.0 mm^2	—
	Panel Aspect Ratio, c/d	—	1.503	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.121	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	1.704	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	69.6 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	550.7 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.24	PASS
G2	Applied Shear, V_u	Governing LC (ULS)	130.99 kN	—
	Shear Area, A_v	$d_w \times t_w$	8704.0 mm^2	—
	Panel Aspect Ratio, c/d	—	1.503	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.121	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	1.704	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	69.6 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	550.7 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.24	PASS
G3	Applied Shear, V_u	Governing LC (ULS)	130.99 kN	—
	Shear Area, A_v	$d_w \times t_w$	8704.0 mm^2	—

	Panel Aspect Ratio, c/d	—	1.503	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.121	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	1.704	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	69.6 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	550.7 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.24	PASS

Note: IS 800 Cl. 8.4, IRC 22 Cl. 603.3.3.2

Table 5.5: Interaction Checks (M-V and M-N)

	Check	Condition	Value	Status
G1	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	4409.64 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.16	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	0.00 + 0.16 = 0.163	0.163	PASS
G2	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	4409.64 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.16	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	0.00 + 0.16 = 0.163	0.163	PASS
G3	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	4409.64 kN-m	—

	Interaction Check: $M_u \leq M_{dv}$	—	0.16	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	0.00 + 0.16 = 0.163	0.163	PASS

Note: IRC 22 Cl. 603.3.3.3

Table 5.6: Lateral Torsional Buckling Check – Construction Stage

	Parameter	Formula	Value	Status
G1	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	12769.32 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.452	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.87	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	2316.41 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.16	PASS
G2	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	12769.32 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.452	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.87	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	2316.41 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.16	PASS
G3	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	12769.32 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.452	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.87	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	2316.41 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.16	PASS

Note: IRC 22 Cl. 603.3.3.1, IS 800 Cl. 8.2.2

Table 5.7: Stiffener Design Summary

G1	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	14.0
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	0.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G2	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	14.0
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	0.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G3	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	14.0
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	0.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None

Table 5.8: End Panel Stiffener Checks

	Check	Required	Provided	Status
G1	Web Buckling Resistance	160.94 kN	109506.26 kN	PASS
	Local Crushing Resistance	160.94 kN	222.03 kN	PASS
	Bearing Capacity	160.94 kN	2100.0 kN	PASS
	Column Buckling Resistance	160.94 kN	1864.55 kN	PASS
G2	Web Buckling Resistance	160.94 kN	109506.26 kN	PASS
	Local Crushing Resistance	160.94 kN	222.03 kN	PASS
	Bearing Capacity	160.94 kN	2100.0 kN	PASS
	Column Buckling Resistance	160.94 kN	1864.55 kN	PASS
G3	Web Buckling Resistance	160.94 kN	109506.26 kN	PASS
	Local Crushing Resistance	160.94 kN	222.03 kN	PASS
	Bearing Capacity	160.94 kN	2100.0 kN	PASS
	Column Buckling Resistance	160.94 kN	1864.55 kN	PASS

Note: IS 800 Cl. 8.4.2.2

Table 5.9: Serviceability – Deflection Checks

	Check	Allowable	Actual	Status
G1	Live Load Deflection, δ_{LL} (mm)	L/800 = 25.0 mm	15.581	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 33.3 mm	37.064	FAIL
G2	Live Load Deflection, δ_{LL} (mm)	L/800 = 25.0 mm	15.611	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 33.3 mm	37.125	FAIL
G3	Live Load Deflection, δ_{LL} (mm)	L/800 = 25.0 mm	15.581	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 33.3 mm	37.064	FAIL

Note: IRC 22 Cl. 604.3.2

Table 5.10: Serviceability – Maximum Stress Limitation

	Element	Allowable Stress	Actual Stress	Status
G1	Structural Steel (0.9 f_y)	315.00 MPa	67.25 MPa	PASS
G2	Structural Steel (0.9 f_y)	315.00 MPa	67.25 MPa	PASS
G3	Structural Steel (0.9 f_y)	315.00 MPa	67.25 MPa	PASS

Table 5.11: Serviceability – Fatigue Assessment

	Stress Range, $\Delta\sigma$ (MPa)	Fatigue Limit, f_{fd} (MPa)	Utilization Ratio	Status
G1	125.03 MPa	92.49 MPa	1.35	FAIL
G2	123.74 MPa	92.49 MPa	1.34	FAIL
G3	123.02 MPa	92.49 MPa	1.33	FAIL

Note: IRC 22 Cl. 605 — governing of normal and shear fatigue (worst by DCR). Capacity reduction factor μ_r applied where plate thickness ¿ 25 mm.

Table 5.12: Girder Design Summary (DCR / Utilization Ratio)

Girder	Controlling LC / Combination	Controlling Check	Demand	Capacity	UR	Status
G1	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	125.03 MPa	92.49 MPa	1.352	FAIL
G2	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	123.74 MPa	92.49 MPa	1.338	FAIL
G3	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	123.02 MPa	92.49 MPa	1.330	FAIL

Note: UR = Demand / Capacity. A value ≤ 1.0 indicates a passing check. The controlling check is the criterion with the highest UR for each girder, with the real load case/combination that drives it.

Table 5.13: Shear Connector Capacity

Parameter	Formula	Value	Reference
Design Resistance, Q_u	$Q_u = \min(Q_{u,s}, Q_{u,c})$ $Q_{u,s} = \dfrac{0.8\, f_u\, (\pi d^2/4)}{\gamma_v}$ $Q_{u,c} = \dfrac{0.29\, \alpha\, d^2 \sqrt{f_{ck}\, E_{cm}}}{\gamma_v}$	95.36 kN	IRC 22 Cl. 606.3.1 (Eq. 6.1)
Fatigue Shear Resistance, Q_r	IRC 22 Table 8 (ϕd, N_{sc})	25.00 kN	IRC 22 Cl. 606.3.2 (Table 8)

Table 5.14: Shear Connector Spacing

Criterion	Governing Spacing	Actual Spacing Provided	Status
ULS Shear (SL1)	1855.1 mm	308.6 mm	PASS
Full Composite (SL2)	308.6 mm	308.6 mm	PASS
SLS Fatigue (SR)	535.7 mm	308.6 mm	PASS
Max Spacing Limit (IRC 22)	400.0 mm	308.6 mm	PASS

Note: IRC 22 Cl. 606.4, 606.9. Governing spacing $= \min(S_{L1}, S_{L2}, S_R)$.

Table 5.15: Transverse Shear and Detailing Checks

Check	Value	Status
Longitudinal Shear per unit length, V_L	102.81 kN/m	—
Transverse Shear Capacity of Slab, V_{Rd}	999.28 kN/m	—
Transverse Shear Check	$V_L/V_{Rd} = 0.10$	PASS
Min. Transverse Reinforcement, $A_{st,min}$	Required 0.51 cm^2/m, Provided 9.44 cm^2/m	PASS
Stud Diameter $\leq 2\,t_f$	$d = 20.0$ mm $\leq 2t_f = 32.0$ mm	PASS
Stud Edge Distance	Provided 110.0 mm (req. $\geq$ 25.0 mm)	PASS

Note: IRC 22 Cl. 606.6, 606.10.

5.2 Deck Slab Design

The reinforced concrete deck slab is designed per IRC 112:2011 (flexure, shear, crack width) and IRC 22:2014 (composite construction). Wheel loads are distributed using Pigeaud's method. The deck is checked for flexure in the transverse and longitudinal directions, punching shear, one-way (beam) shear, crack width, and reinforcement detailing.

Table 5.16: Deck Slab — Loading and Geometry

Effective Span of Deck Slab, l_{eff}	1717 mm (girder spacing, c/c)
Deck Thickness, t_s	250.0 mm
Clear Cover (IRC 112 Cl. 15.2)	Top 50 / Bottom 50 mm
Concrete Grade (IRC 112 Cl. 6.4)	M40 (f_{ck} = 40.0 MPa, f_{ctm} = 3.0 MPa)
Reinforcement Grade (IRC 112 Cl. 6.2)	Fe 500 (f_y = 500 MPa)
Dead Load per Unit Area, w_{DL}	6.13 kN/m^2 (slab self-weight)
IRC 6 Wheel Load (Class A / 70R)	55.9 kN
Tyre Contact Width (IRC 6 Annex A)	250 mm (transverse)
Impact Factor (IRC 6 Cl. 208.2)	1.269
Governing Live Load Case	ClassA

Table 5.17: Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method)

Location	Parameter	Formula / Reference	Value	Status
At Midspan (Sagging)	Transverse BM (DL), $M_{T,DL}$	$w_{DL}\, l_{eff}^2/10$	1.81 kN-m/m	—
	Transverse BM (LL), $M_{T,LL}$	Effective width (IRC 112 B3.1)	16.37 kN-m/m	—
	Total Design BM, $M_{u,sag}$	1.35 DL + 1.50 LL	33.60 kN-m/m	—
	Effective depth, d	$t_s - c_{nom} - \phi/2$	194.0 mm	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	37.05 kN-m/m	PASS
At Support (Hogging)	Total Design BM, $M_{u,hog}$	1.35 DL + 1.50 LL (at support)	25.20 kN-m/m	—
	Required Top Steel, $A_{st,top}$	$M_u/(0.87\, f_y\, d)$	488 mm^2/m	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	40.16 kN-m/m	PASS

Note: IRC 112 Cl. 12.2. Distribution (longitudinal) reinforcement designed for 20% of main steel moment (IRC 21 Cl. 305.18).

Table 5.18: Deck Slab — Cantilever Overhang Flexure Check

Parameter	Formula	Value	Status
Overhang Length, l_{oh}	—	0.8583 m	—
Crash Barrier Load Moment	IRC 6 Cl. 206.4	7.50 kN-m/m	—
Dead Load Moment	$w_{DL}\, l_{oh}^2/2$ + railing	3.52 kN-m/m	—
Live Load Moment (eccentric wheel)	Wheel load × arm	33.01 kN-m/m	—
Total Hogging Moment, $M_{u,oh}$	1.35 DL + 1.50 (LL + CB)	78.83 kN-m/m	—
Moment Capacity (top steel), $M_{Rd,oh}$	IRC 112 Cl. 12.2	84.49 kN-m/m	PASS

Note: IRC 6 Cl. 206.4 crash barrier loads applied at kerb face; IRC 112 Cl. 12.2 flexure.

Table 5.19: Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6)

Parameter	**Formula / Reference**	**Value**	**Status**
Design Wheel Load (ULS), V_{Ed}	$\gamma_Q\,(1 + IF)\,P_w$	106.4 kN	—
Tyre Contact Area	$a \times b$ (IRC 6 Annex A)	250 × 200 mm	—
Loaded Area at mid-depth, b_0	$c_1 \times c_2$ (incl. WC dispersion)	350 × 300 mm	—
Control Perimeter, u_1	$2(c_1 + c_2) + 4\pi d$	3738 mm	—
Punching Shear Stress, v_{Ed}	$V_{Ed}/(u_1\,d)$	0.147 MPa	—
Punching Resistance, $v_{Rd,c}$	IRC 112 Eq. 10.1	0.555 MPa	—
Punching Shear Check	$v_{Ed} \leq v_{Rd,c}$	0.26	PASS

Note: Punching shear reinforcement not typically required for deck slabs with $d \geq 200$ mm and adequate longitudinal reinforcement.

Table 5.20: Crack Width Check (Deck Slab)

Parameter	Value / Reference
Min. Reinforcement for Crack Control, $A_{s,min}$	303 mm²/m [IRC 112 Cl. 16.5.1]
Provided Reinforcement (bottom)	ϕ12 @ 250 mm c/c (452 mm²/m)
Max. Permissible Crack Width	0.30 mm
Calculated Crack Width, w_k (governing)	0.2314 mm
Crack Width Check	PASS

Table 5.21: One-Way (Beam) Shear Check (Deck Slab)

Parameter	Formula / Reference	Value	Status
Design Shear per unit width, V_{Ed}	$\gamma_{DL}V_{DL} + \gamma_{LL}(1+IF)V_{LL}$	79.72 kN/m	—
Effective depth, d	$t_s - c_{nom} - \phi/2$	194.0 mm	—
Size factor, k	$1 + \sqrt{200/d} \leq 2.0$	2.000	—
Long. reinforcement ratio, ρ_l	$A_{sl}/(b_w\,d) \leq 0.02$	0.0023	—
Shear resistance (no stirrups), $V_{Rd,c}$	$v_{Rd,c}\,b_w\,d$ (Cl. 10.3.2)	107.58 kN/m	—
One-Way Shear Check	$V_{Ed} \leq V_{Rd,c}$	0.74	PASS

Note: IRC 112 Cl. 10.3.2. Shear reinforcement not provided in deck slabs; capacity relies on concrete and main reinforcement.

Table 5.22: Reinforcement Detailing Summary (Deck Slab)

Parameter	Required / Limit	Provided	Status
Main Reinforcement — Bottom (Transverse)			
Required Area, $A_{st,req}$ (mm²/m)	431 mm²/m	452 mm²/m	PASS
Bar Diameter × Spacing	$\phi \geq 10$ mm (IRC 112)	ϕ12 @ 250 mm c/c	—
Min. Reinforcement $A_{s,min}$ (IRC 112 Cl. 16.3.1)	303 mm²/m	452 mm²/m	PASS
Max. Bar Spacing (IRC 112 Cl. 16.3.2)	300 mm	250 mm	PASS
Distribution Reinforcement — Longitudinal			
Required Area, $A_{st,dist}$ (mm²/m)	$\geq$ 20% of main steel	377 mm²/m	PASS
Top Reinforcement (Support / Cantilever Overhang)			
Required Area, $A_{st,top}$ (mm²/m)	488 mm²/m	492 mm²/m	PASS
Cover and Detailing			
Clear Cover (IRC 112 Cl. 15.2)	$\geq$ 50 mm (Table 14.2)	Top 50 / Bottom 50 mm	PASS

Note: IRC 112 Cl. 16.3, IS 456 Cl. 26.5. All reinforcement provisions satisfy strength and detailing requirements.

5.3 Cross Bracing Design

Cross bracing between adjacent plate girders provides lateral stability during construction, resists transverse loads (wind, seismic, braking) in service, and prevents lateral torsional buckling of the girders. Members are designed per IS 800:2007 Cl. 7 (compression) and Cl. 6 (tension). Forces are derived from the grillage model under the governing load combination (DL + LL + WL).

Table 5.23: Cross Bracing — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
G1-G2	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G1-G2	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.24: Cross Bracing — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
G1-G2	Diagonal	C	1963	103.7	250 — PASS
	Top chord	C	1717	90.7	250 — PASS
	Bottom chord	T	1717	90.7	400 — PASS
G2-G3	Diagonal	C	1963	103.7	250 — PASS
	Top chord	C	1717	90.7	250 — PASS
	Bottom chord	T	1717	90.7	400 — PASS

Note: 3. Limit = 250 for compression members, 400 for tension members. $K = 1.0$ for members with both ends pinned.

Table 5.25: Cross Bracing Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	40 x 40 x 3	1.5 EQ (b)	2.915	24.220	0.270	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	2.549	28.430	0.230	PASS
G2-G3	Diagonal	40 x 40 x 3	1.5 EQ (b)	3.446	45.000	0.080	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	3.013	45.000	0.070	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.4 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.

Table 5.26: End Diaphragm — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm^2)	r_{min} (mm)
G1-G2	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G1-G2	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.27: End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
G1-G2	Diagonal	C	1963	103.7	250 — PASS
	Top chord	C	1717	90.7	250 — PASS
	Bottom chord	T	1717	90.7	400 — PASS
G2-G3	Diagonal	C	1963	103.7	250 — PASS
	Top chord	C	1717	90.7	250 — PASS
	Bottom chord	T	1717	90.7	400 — PASS

Note: 3. Limit = 250 for compression members, 400 for tension members. $K = 1.0$ *for members with both ends pinned.*

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	40 x 40 x 3	1.5 EQ (b)	2.915	24.220	0.270	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	2.549	28.430	0.230	PASS
G2-G3	Diagonal	40 x 40 x 3	1.5 EQ (b)	3.446	45.000	0.080	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	3.013	45.000	0.070	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.5 Overall Design Check Summary

Table 5.29: Overall Design Check Summary — All Members

Member / Check	Governing Load Combo	Demand	Capacity	UR
Girder — Moment	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	697.45 kNm	4409.64 kNm	0.16
Girder — Shear	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	160.94 kN	550.70 kN	0.29
Girder — LTB (constr.)	1.0 DL	365.92 kNm	2316.41 kNm	0.16
Girder — Deflection	1.0 DL + 1.0 LL	13.69 mm	33.33 mm	0.41
Girder — Stress	SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	68.94 MPa	315.00 MPa	0.22
Girder — Fatigue	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	125.03 MPa	92.49 MPa	1.35

Transverse Shear (slab)	—	—	—	—
Crack Width (slab)	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	0.231 mm	0.300 mm	0.77
Deck — Flexure (sagging)	Basic ULS: 1.35DL + 1.5LL	33.60 kN-m/m	37.05 kN-m/m	0.91
Deck — Flexure (hogging)	Basic ULS: 1.35DL + 1.5LL	25.20 kN-m/m	40.16 kN-m/m	0.63
Deck — Cantilever Overhang	Basic ULS: 1.35DL + 1.5LL	78.83 kN-m/m	84.49 kN-m/m	0.93
Deck — Punching Shear	Basic ULS: 1.35DL + 1.5LL	0.15 MPa	0.55 MPa	0.26
Deck — One-Way Shear	Basic ULS: 1.35DL + 1.5LL	79.72 kN/m	107.58 kN/m	0.74
Cross Bracing — Compression	1.5 EQ (b)	6.54 kN	24.220 kN	0.27
Cross Bracing — Tension	1.5 EQ (b)	3.60 kN	45.000 kN	0.08
Cross Bracing — Slenderness	—	103.7	250	0.41
End Diaphragm — Moment	1.5 EQ (b)	6.54 kN	24.220 kN	0.27
End Diaphragm — Shear	1.5 EQ (b)	3.60 kN	45.000 kN	0.08

Note: UR = Demand / Capacity. All values ≤ 1.0 *indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.*

6 Drawings and Visualizations

This section presents CAD-generated views of the designed bridge and its components. All views are generated automatically by OsdagBridge using pythonOCC.

6.1 Bridge Configuration and Layout

[PLACEHOLDER: Overall 3D Bridge Superstructure]

Figure 6.1: Overall 3D Bridge Superstructure

[PLACEHOLDER: Typical Cross Section]

Figure 6.2: Typical Cross Section

[PLACEHOLDER: Top View]

Figure 6.3: Top View

6.2 Plate Girder — Detailed Views

[PLACEHOLDER: 3D View of Plate Girders]

Figure 6.4: 3D View of Plate Girders

[PLACEHOLDER: Cross Section of Plate Girder]

Figure 6.5: Cross Section of Plate Girder

[PLACEHOLDER: Side View of Girder]

Figure 6.6: Side View of Girder

6.3 Cross Bracing Detail

[PLACEHOLDER: Cross Bracing Layout]

Figure 6.7: Cross Bracing Layout

[Cross Bracing – Bracing Section]

Figure 6.8: Cross Bracing – Bracing Section

[Cross Bracing – Top Chord Section]

Figure 6.9: Cross Bracing – Top Chord Section

[Cross Bracing – Bottom Chord Section]

Figure 6.10: Cross Bracing – Bottom Chord Section

6.4 End Diaphragm Detail

[PLACEHOLDER: End Diaphragm Layout]

Figure 6.11: End Diaphragm Layout

[End Diaphragm – Bracing Section]

Figure 6.12: End Diaphragm – Bracing Section

[End Diaphragm – Top Chord Section]

Figure 6.13: End Diaphragm – Top Chord Section

[End Diaphragm – Bottom Chord Section]

Figure 6.14: End Diaphragm – Bottom Chord Section

7 Material Take-off & Quantity Summary

Table 7.1 Bill of Materials (Steel, Concrete, and Reinforcement Quantities)

S.N.	Item Description	Volume	Quantity	Total Volume	Weight (MT)	Total Weight (MT)
1	Structural Steel (IS 2062) for Girders	$0.01559\ m^2 \times 20.00\ m = 0.31184\ m^3$	3	0.94	2.45	7.34
2(a)	Cross Bracing - Top Chord	$0.00024\ m^2 \times 1.72\ m = 0.00041\ m^3$	12	0.00	0.0032	0.04
2(b)	Cross Bracing - Bottom Chord	$0.00024\ m^2 \times 1.72\ m = 0.00041\ m^3$	12	0.00	0.0032	0.04
2(c)	Cross Bracing - Diagonal Chord	$0.00024\ m^2 \times 1.96\ m = 0.00047\ m^3$	24	0.01	0.0037	0.09
3	Concrete (M40) for Deck Slab	$5.15\ m \times 0.25\ m \times 20.00\ m = 25.75\ m^3$	1	25.75	64.38	64.38
4	Reinforcement Steel (Fe 500)	$0.019682\ m^2 \times 20.00\ m = 0.39363\ m^3$	1	0.39	3.09	3.09
5	Shear Stud Connectors	$0.000314\ m^2 \times 0.100\ m = 0.000031\ m^3$	390	0.01	0.000247	0.096
6	Crash Barrier	$0.00000\ m^2 \times 20.00\ m = 0.00001\ m^3$	2	0.00	0.00	0.00

8 Standards & Assumptions

This section provides references to standards used to calibrate the OsdagBridge design modules, and notes the limitations of the current software version.

8.1 Design Standards

The following Indian Road Congress (IRC) codes and Indian Standards (IS) form the basis of all design calculations in this software.

Table 8.1: IRC Codes

Code	Year	Title / Scope
IRC 5	2015	General Features of Design - carriageway widths, kerb, footpath dimensions
IRC 6	2017	Loads and Load Combinations - dead load, live load, impact, wind, temperature, etc.
IRC 22	2015	Composite Construction (LS) - Composite section properties, ULS/SLS design, shear connectors
IRC 24	2010	Steel Road Bridges (LS) - Stiffener design, skew angle limits, diaphragm requirements
IRC 112	2020	Concrete Road Bridges - Deck slab flexure, shear, crack width, reinforcement
IRC SP 114	2018	Seismic Design of Road Bridges

Table 8.2: IS Codes

Code	Year	Scope
IS 800	2007	Steel construction - tension, compression, bending, shear, LTB, stiffeners, combined checks
IS 456	2000	Concrete - simplified stress-block for deck moment capacity
IS 1786	2008	Reinforcement steel properties
IS 1893 (Part 3)	2014	Earthquake resistant design
IS 2062	2011	Structural steel - yield and ultimate strength by grade

8.2 Analysis and Design Assumptions of This Version

Structural Analysis

- All girders are modelled as simply supported; continuous spans are not currently supported.
- A 3D grillage model (OSPGrillage) is used for load distribution. Grillage members carry composite section properties after the construction stage.
- The transverse member forces are computed using an approximate 2D frame analogy. For irregular or skewed geometries, a 3D FEM is recommended.
- Fixed bearing stiffness is modelled as $k = 1{,}000{,}000\,\mathrm{kN/m}$ (virtually rigid); free/expansion bearing as $k = 100\,\mathrm{kN/m}$.
- Construction stage sequence analysis is approximate. Detailed staged analysis should be performed for long-term deflection checks.

Composite Action

- Full shear connection is assumed at ULS with headed stud connectors designed per IRC 22:2015 Cl.606.
- Short-term composite section properties (modular ratio $n = E_s/E_{cm}$) are used for ULS checks.
- Long-term composite section properties (with creep-adjusted modular ratio) are used for SLS deflection and crack-width checks.
- Pre-composite stage: the steel girder alone resists all construction loads prior to concrete gaining strength.

Material Properties (IRC 22:2015 Annex III)

- Steel: $E_s = 200{,}000\,\mathrm{MPa}$, $G_s = 80{,}000\,\mathrm{MPa}$, $\nu = 0.30$, $\alpha = 11.7 \times 10^{-6}/^{\circ}\mathrm{C}$ (grade-independent).
- Minimum structural concrete grade: M25.
- Default reinforcement grade: Fe500.

Partial Safety Factors (IRC 22:2015 Cl.601.4)

- γ_{m0} (steel yield, ULS) $= 1.10$
- γ_{m1} (steel ultimate, ULS) $= 1.25$

- Reinforcement (ULS) = 1.15
- Welds – shop: 1.25; field: 1.50
- Fatigue (γ_{mft}) = 1.35

Loading

- Dead load densities per IRC 6:2017 Cl.203: structural steel 78.5 kN/m^3, concrete 25.0 kN/m^3, bituminous wearing course 24.0 kN/m^3.
- Multi-lane live load reduction factors per IRC 6:2017 Cl.204.4 Table 6A: 1st lane = 1.0, 2nd lane = 0.8, 3rd lane onwards = 0.4.
- Impact factor (dynamic load allowance) computed from span per IRC 6:2017 Cl.208.2/208.3.
- Wind load applied as transverse, longitudinal, and vertical components per IRC 6:2017 Cl.209.3.3–209.3.5.

Serviceability Limits

- Deflection limits (IRC 22:2015 Cl.604.3.2): live load + impact $\leq L/800$; total $\leq L/600$.
- SLS stress limits: concrete $\sigma_c \leq 0.48 f_{ck}$; rebar $\sigma_s \leq 0.80 f_{yk}$; steel $f_e \leq 0.9 f_y$.
- Permissible crack width: $w_k \leq 0.3$ mm (bridge deck, exposure class XS2/XD2 per IRC 112:2020 Cl.12.3.2).

Fatigue

- Reference fatigue life: $N_{sc} = 2 \times 10^6$ cycles (IRC 22:2015 Cl.605).
- Constant stress range is assumed. A thickness correction factor μ_r is applied for plate thickness > 25 mm.

Stiffener Design

- Intermediate transverse stiffener and bearing stiffener design follows IS 800:2007 Cl.8.7.2/8.7.3 and IRC 24:2010 Cl.509.7.2/509.7.3.

8.3 Known Limitations of This Version

- Substructure (piers, pile caps, foundations) and bearing design are not included.
- Splice connection design is not implemented.
- Skew angle > 15 degrees requires independent manual analysis (IRC 24 Cl. 504.8).
- Construction stage sequence analysis is approximate; detailed staged analysis should be performed for long-term deflection checks.
- The grillage analysis assumes simply supported boundary conditions; continuous spans are not currently supported.

References

1. IRC 5 (2024) — *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*
2. IRC 6 (2017) — *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations.*
3. IRC 22 (2014) — *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*
4. IRC 24 (2010) — *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*
5. IRC 112 (2011) — *Code of Practice for Concrete Road Bridges.*
6. IRC SP 114 (2018) — *Guidelines for Seismic Design of Road Bridges.*
7. IS 800 (2007) — *Indian Standard: General Construction in Steel — Code of Practice.*
8. IS 2062 (2011) — *Hot Rolled Medium and High Tensile Structural Steel — Specification.*
9. Subramanian, N. (2008). *Design of Steel Structures.* Oxford University Press.
10. Steel-INSDAG Teaching Resource Materials. `https://www.steel-insdag.org`